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Knowledge sharing in practice: a game-based methodology to increase farmers' engagement in a common vision for a cheese PDO union

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ABSTRACT

Purpose: Knowledge Sharing (KS) is seen as a promising framework for renewing advisory systems in agriculture, but its practical application remains an important area of research. In particular, it is important to increase the motivation of farmers to participate in decision-making. Game-based support seems to be applicable to a long-term process of KS.

Approach: The authors spent one year developing a methodology to help identify a common vision among multiple stakeholders in a PDO area. This area of cheese production (Fourme de Montbrison) faces important difficulties with sustainability issues and a very low degree of farmer participation. The methodology is based on game learning and incorporates three phases and six steps and is centred on two serious games.

Findings: Results present an analysis of the farmers' participation and the common vision for the future that the game-based methodology enabled us to identify. The games facilitated the farmers' implication in collective decision-making and their continued engagement afterwards.

Practical implications: the article discusses the advantages of the game-based methodology, such as involving local knowledge of farmers at all stages and encouraging their long-term engagement in collective decision-making.

Theoretical implications: the article complements the existing scientific literature on KS by emphasizing the importance of long-term perspectives in a game-based learning approach. From a socio-constructivist perspective, it also highlights the potential of game-based learning to motivate farmers in collective decision-making.

Originality/value: This support methodology could be a source of inspiration for extension actors facing similar development problems and the revitalization of their role as facilitators.

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Introduction

The variety of food and agriculture models (conventional, modern, ecological intensive, etc.) and their coexistence within the same sector and/or region (IPES 2016; Cayre et al. 2018; Plumecocq et al. 2018) lead us to reconsider the learning and action processes between actors, especially since top-down methods have several limits. As discussed by Coquil et al. (2018), interventions based on generic knowledge are particularly ill-adapted to the growing expression of local particularities (ecological processes, micro-climates, etc.) and to the singularities of farmers. Cayre et al. (2018) illustrated these singularities among the contrasting world-views on nature and modernization expressed by farmers from three different areas of French PDO (Protected Designation of Origin) cheese production. Consequently, we are currently witnessing a renovation of advisory methods, moving away from the postures of supervision and dissemination of technical knowledge towards co-design and participatory approaches (Knickel et al. 2009). In these new methods, collaboration between actors is strongly advocated (Dogliotti et al. 2014; Lacombe, Couix, and Hazard 2018) to promote non-academic (Reed 2008) and situational knowledge (Coolsaet 2016; Toffolini et al. 2017). The concept of knowledge sharing (KS) in agriculture, has a long history among farmers (Chambers and Ghildyal 1985) and has been increasingly seen as significant (Roling and Wagemakers 2000; Mitton et al. 2007; Raymond et al. 2010). We view KS here as the valorization of local, non-scientific knowledge, that is anchored in the collective that produces it (Goulet 2013; Dogliotti et al. 2014; Methorst et al. 2017; Hazard et al. 2018).

As Lyet (2018) points out, the aim of KS is not to attribute a principle of truth to all knowledge produced, but rather to enable and encourage individuals to appropriate knowledge through sharing and debate. It is also about reaching out to people who are usually excluded from the debate because they are outside of the traditional circle of political or administrative representatives (Coudel, Tonneau, and Rey-Valette 2011). The operationalization of KS in practice, and its application in decision-making, remain a crucial area of research. In particular, Taylor and de Loë (2012) show that scientists' and public managers' epistemological anxiety about local knowledge was a significant barrier to its effective use in decision-making. The linear view of knowledge transfer, as a top-down process from research to advice to farmers, is still predominant (Moschitz et al. 2015), but it fails to engage farmers on ecological issues (De Snoo et al. 2013) and does not recognize their progress and innovation (Burton and Paragahawewa 2011). Moreover, local knowledge holders (farmers for example) do not always feel concerned, qualified or competent to participate in the governance of their sector (Sterling et al. 2017; Fielke et al. 2018). Over time, left out of the deliberative process, the farmers reduce their participation (Berthet and Hickey 2018; Dolinska et al. 2020). This situation is even more evident when local economic norms and institutions induce or increase competition and individualism among community members (Gómez-Baggethun et al. 2010; Koutsou, Partalidou, and Ragkos 2014).

Game-based learning (Abt 1970) approaches to KS seem to be a promising response to this issue, in part because they promote trust between participants (Plass, Homer, and Kinzer 2015; Berthet et al. 2016). In recent years, numerous agricultural research studies have promoted games as a major tool for engaging farmers (Dolinska 2017; García-Barrios et al. 2017; Bosma et al. 2019; Andreotti et al. 2020; Hernandez-Aguilera

et al. 2020). However, mobilizing games poses a problem: it is an approach that is often focused on short-term projects (one year or less), that do not take account of situations where a long-term engagement in decision-making is needed, as is the case for farmers (Flood et al. 2018; Rodela, Ligtenberg, and Bosma 2019). In a broader sense, this problem is observed throughout the game-based learning literature (Girard, Ecalle, and Magnan 2013; Johnson et al. 2017; Madani, Pierce, and Mirchi 2017; Stanitsas, Kirytopoulos, and Vareilles 2019) and also in participation process literature (Felt et al. 2012; Oliver et al. 2012; Koutsou, Partalidou, and Ragkos 2014; Bandewar et al. 2017; Ernst 2019). The objective of this study was, therefore, to question how the operationalization of KS over the long term can be achieved by maintaining the motivation of farmers that have often been left outside the scope of decision-making in collective deliberation. In this paper, we examine the integration of game-based support as a way to promote the common participation of actors within a long-term process.

From this perspective, this article proposes an original methodology through a game-based approach. This methodology was implemented in 2018 with multiple stakeholders in a French PDO cheese area (la Fourme de Montbrison in central France). This work was initiated at the request of leaders from the PDO union, who wanted to develop a common vision for the future, overcome dissention in the area and a lack of perspectives, and especially address the low engagement of farmers in decision-making. In this paper, we first present the context of the PDO area and the three phases, six steps, and associated tools of the game-based learning methodology for KS. The results section presents the common vision for the future that emerged from this support process, intermediary results that reveal actual KS in the process and post support engagement of farmers. Main outputs from this collective learning process are discussed in the final section with the implications for its application to other cases.

Context

Research on PDO labels for cheeses shows that they generate significant economic value at an EU level, especially in certain countries like France (Jeanneaux 2018). In market-based ecological modernization, production is centred upon the specific qualities of products and distinctive features of an area or region where the various producers come together to form close networks (Maye et al. 2016). Some PDOs may be in difficulty if they are not well known or if their product or region suffers from a poor environmental image, as for example in the Massif-Central of France (Callois et al. 2019). Kneafsey (2010) explains that in this case, the proximity between actors encourages the expression of multiple and sometimes contradictory visions for the products (expectations for the economy, production processes, etc.).

The Fourme de Montbrison cheese is an exemplary case of this situation. A blue cheese produced from cow milk, it is considered a minor appellation in the French cheese PDO field (46 cheeses), both in terms of production levels, geographical scope, reputation and the number of breeders and dairies involved. Currently, the geographical area of the Fourme de Montbrison encompasses 33 villages in the Forez mountains, a region where the altitude ranges from 600 m to 1300 m. When our work began with this PDO, sixty-nine dairy farms with 115 farmers delivered 13.2 million litres of milk in 2018, for the production of some 564 tonnes of cheese by four processors. Two

transformers share more than 90% of the Fourme de Montbrison production. These are large agri-food companies affiliated with two major French groups. The other two processors have a so-called artisanal production. The PDO union is organized in the form of a Board of Directors (BoD) of ten people: five seats for milk producers, five seats for processors. All the members of the union (farmers and processors) elect the directors during a general assembly. A president is chosen and nominated by the directors to represent the union and to apply the policies determined by the BoD. Since the formation of the union, the president has always been a farmer.

Like several other PDOs in this region (Callois et al. 2019), Fourme de Montbrison faces difficulties: a low milk prices, and little control over production volume among farmers prevents the stabilization of distribution (Vollet et al. 2018). In addition, the territory of the Fourme de Montbrison PDO designation has clearly suffered from the turbulence of recent years (bankruptcy of a producers' cooperative, sanitary problems, and emergence of a new cheese in the zone which confused consumers). In the end, the very identity of the product is questioned, the structure of the sector is weakened and, as in several dairy PDOs, an increasing number of dairy farmers are converting their farm to meat production, which is less labour intensive. In the long term, the volume of milk intended for the production of Fourme is threatened, as is the existence of the PDO. However, the recent reshuffling of the members of the BoD of the PDO union, and the stakeholders' strong desire for change are major assets for the future.

As in other dairy sectors, milk price is a central question in the discussions between stakeholders, and the subject risks being a threat to any collaborative process. Recent French PDO laws have abolished certain trade arenas, leading to a fear that the absence of an identified place of exchange could be a constraint to the producers' and processors' joint construction of the product image. In addition, the global economic, regulatory and climatic context places farmers, processors and their partners in positions of high uncertainty (Torre and Vollet 2016). Under these conditions, each stakeholder explored his own adaptive pathway, according to his aspirations and constraints, often in an individual way and usually to the detriment of the collective (not taking into account the collective issues of the PDO). The collective (farmers and processors) thus found itself with a diversity of objectives and points of view that lead to tension when faced with the need to rethink the PDO specifications. For these reasons, farmers' participation in the various PDO assemblies (BoD nominations, working groups) was very low (less than ten active farms in the union) and motivation to invest in the future has suffered. These conditions threaten the long-term viability of this collective structure.

Material and methods

We developed an iterative methodology based on three phases (pre-game, game, post-game) and six steps, where game-based learning had a pivotal role. As in the work of Eastwood, Chapman, and Paine (2012), this promoted iterative exchange, through repeated interaction. The game is effective when it is supplemented with other methods, multiple times in the process, and when players work in groups (Wouters et al. 2013). It was conducted during 2018, as summarized in Figure 1.



Figure 1. Infographics summarizing the support process.

Chosen games

Two games were specifically used for this process, for their ability to meet the challenges of foresight in territorial and livestock farming, and also for their potential to include farmers in the exchange and foster dialogue by taking account of their knowledge. An important objective was to counter the inherent tensions traditionally experienced during interactions in agriculture projects: discourse-dependency, context-dependency, and power asymmetries (Foran et al. 2014). To accomplish this, the games were created in the lab of the main co-authors and incorporated a Vygotskian perspective (Vygotsky 1978) in the design. Learning happens through interacting with others and the symbolic tools of one''s community: game-play is seen as an efficient device to motivate learning among the actors and engage them in a group construction over the long-term (Wu et al. 2012; Plass, Homer, and Kinzer 2015; Qian and Clark 2016). For this, games must incorporate social interactions and dialogue among the learners (Foko and Amory 2008). In a process of decentration, this provides group members with opportunities to integrate their prior experiences in different contexts, mixed with each individual''s knowledge. In this perspective, our games aim at the development of a socio-cognitive conflict (Johnson and Johnson 2009): The players learn by confronting their own representations with those of other players. They are then led to expand their own representations, which reinforces their attraction for the interaction and the support process (Dernat et al. 2018).

The 'Jeu de territoire®' (Angeon and Lardon 2008) is a serious game that aims to build a common vision for the future resolution of a problem with a spatial dimension. This tool integrates multiple stakeholder perspectives. It focusses on farmers' practices and on technical and organizational innovations, while remaining open to the problems of the PDO area and to the current challenges facing society with regard to agriculture and agri-food (agroecology, animal welfare, local food, etc.). For this reason, it can answer the issue of providing holistic and systemic views of problems (Turner et al. 2016). The game has already proven to be useful on numerous occasions during twelve years of use and ameliorations (Soulard and Lardon 2019).

The serious game of 'La Grange®' provides an operational but non-spatialized representation of a livestock farming area (land use, animals, upstream and downstream agri-food industries) as a socio-ecological system (Ryschawy et al. 2019). It focusses on how livestock farming interacts with its physical, economic and social environment, and allows identifying synergies, trade-offs and coordination among its main features and stakeholders (Dumont et al. 2019).

Both games have the advantage of not proposing any pre-established prescriptions or directing exchanges on their possible conclusions towards a 'reference' project (Dalmais, Gasselin, and Cerf 2015). However, they can support KS and the development of critical thinking (Bosma et al. 2019).

Pre-game phase

The first step was to present and share the methodology (all the steps and phases presented here) with farmers and processors. This step aimed to clarify and discuss the timetable with the participants, taking into account work arrangements, the local context (production constraints, local representations of farmers), and the objectives both for

the Fourme union and for researchers. Two meetings were held in February and March. Nearly 60% of farms ($n = 40$) and all four processors participated. This enabled the entire support process to be set in motion, adapted to the constraints of each individual, and prepared for the second task. The second step consisted of collecting information from the actors that formed both an inventory of the current situation and a forecast for that in the future. This data was necessary for the game design and is traditionally used for building games. Interviews were conducted directly on the farms and in dairies. The interviews took place from March to July. About thirty farms and processors were interviewed for an average of two hours. All the interviews were then treated by a thematic analysis and the construction of a mind map grouping all the themes and their elements (Chemel 2018). This step also involved additional data collected from a local survey (in local markets) conducted with Fourme consumers (Mauduit 2018). This data was integrated into the context elements of the design of both games.

Game phase

The two serious games were employed during a collaborative day bringing together AOP actors and their partners. Eighty-nine stakeholders were present, including about 40 farmers and all the representatives of the 4 processors. There were also local public decision-makers and government representatives as well as education (teachers), animal health (veterinarians), agriculture (advisors) and tourism stakeholders. All farmers and processors were invited. The external stakeholders present were selected and contacted by the researchers from among the major local actors involved in the development of the PDO. Traditionally mobilized in the two chosen games, they provide the different points of view and add new knowledge. The external stakeholders thus act as mediators from the socio-constructivist perspective of support by questioning farmers and processors with regard to their own issues. The games were played in parallel sessions in small groups of 6–8 people arranged by tables (Figures 2 and 3) incorporating a mix of different actors (farmers, processors and other stakeholders). Each actor could take the floor and contribute to the shared worldview and a common vision for the future. At the end of the day, the proposals for various positions identified were pooled, shared and discussed in order to provide a first framework for a roadmap. This debriefing phase is very important in the game-based process (Peters and Vissers 2004). The animation of each table was carried out by a research staff member previously trained by the co-authors. In addition, they each received an operational kit containing the instructions. The training and the kit focused on strategic dimensions of learning through play inspired by socio-constructivist literature: peaceful management of exchanges, management of role asymmetries, management of speech, valorization of local knowledge, personal experiences and anecdotes (Dernat et al. 2018).

Post-game phase

The next step of the methodology (step 4) was a vote to prepare the multi-year collective action plan. This was prepared on the basis of the results of the collaborative day (i.e. the different actions proposed by participants in the games). These actions were submitted to the votes of the PDO stakeholders (farmers and processors) during an online survey



Figure 2. A table of the Jeu de Territoire during the collaborative day.

(LimeSurvey) in November 2018. More than 70 responses were collected. The aim was to prioritize the collective''s expectations and to identify aspirations for the future. The data from the responses were compiled and discussed at two meetings of the BoD of the PDO union, with the participation of the researchers (step 5). These two days of reflection led the Board members to define a proposal for a 10-year strategy based on the results of the



Figure 3. A table of La Grange during the collaborative day.

entire process (data collection phase, collaborative day, online survey). In February 2019 the strategy was presented at a meeting devoted specifically to the farmers and processors of the PDO union (step 6). This meeting allowed individual points of view to be expressed while strengthening the collective sharing of results and choices made. Then in March 2019 an ordinary general assembly together with the partners (elected officials and State actors in particular) approved the strategy.

Results

Engagement of farmers in different steps of the process

The different levels of participation of farmers in the different steps are visible in Table 1. Farmers' engagement is expressed in terms of the presence of at least one farmer from the farm (some farms having several associated farmers). Thus, the number of farmers present was sometimes higher (as on the game day), but in our opinion the number of participating farms is a clearer measure of the engagement. Throughout the steps, farm participation remains high. We believe that the low participation during the data collection step is due to its time-consuming nature. In fact, meeting with all farmers was impossible because it would have required an inordinate amount time and resources. Nevertheless, we also believe that the farms (half of the total) in this step gave a representative sense of the farmers' local knowledge and expectations.

Table 1. Farm engagement in different steps of the process.

Steps	Methodology	Farms participating	Results
Pre-game phase			
1	Meetings	40 / 69	Engagement of farmers
2	Data collection and integration in games	28 / 69	Local knowledge of farmers
Game phase			
3	Territory game & La Grange	40 / 69	53 actions co-constructed with farmers
Post-game phase			
4	Electronic vote	55 / 69	Actions prioritised with farmers
6	Meetings	46 / 69	Common strategy for PDO union

We did note the recurrent absence of farmers from about 20 farms. These were always the same farms. It was never possible to assemble all of them in the process, despite reminders. Many farmers were reluctant, or not interested. Several explanations and arguments were collected. The predominant reason for non-participation was due to the age of some farmers who were soon to retire, without a successor. This fact limited their interest to participate in a process for the future of a PDO in which they would not or no longer be involved. Some, a minority, were also against the process because of their history with the PDO union (as explained in context: tensions, uncertainties, etc.). Finally, a significant number never responded to our requests. We suspect that these were also related to a problem with the PDO union.

No refusal to participate was, to our knowledge, motivated by the approach focused on games, nor the presence of researchers. On the contrary, the mobilization of the games, although potentially confusing, appeared to reassure many of participants about their ability to express themselves freely.

Co-constructed actions

The 53 actions that emerged during the game collaborative day and were then submitted to the vote are presented in Table 2, together with the number of farmers and processors who voted for each action (participation), the average score and standard deviation. Some actions specify the desire to preserve production methods that refuse certain forms of modernization (Cayre et al. 2018), for instance, the prohibition of milking robots. Others refer to forms of production and economic strategies that compromise the notion of PDO (purchase of fodder outside designated area), or to more innovative marketing ideas (creation of a cheese bar). We also see the desire to return to more traditional forms of agriculture, such as larger amount of hay in winter rations to reduce inputs. Participation, average score, and standard deviation, show that stakeholders sometimes reached agreement on actions that initially lacked consensus. This is the case, for example, for actions relating to the negotiation of milk prices or to the image of raw milk. These actions were initially proposed by only one table of participants during the gaming day.

A constructed vision for the future

After negotiation in the PDO Board meetings, the members finally adopted guidelines that constitute the backbone of a new common vision for the future. These guidelines

Table 2. Actions proposed during the collaborative day (step 3), votes on these actions (step 4) and final guidelines relevant (step 5).

Proposed actions	Note/ 5[a]	Standard deviation	Participation[b]	Guidelines
Action 13: Improve the image of raw milk.	3.88	1.35	18	1
Action 26: Organize open house days at PDO farms and processors.	3.48	1.33	16	1
Action 08: Promote *le Forez* (the local area).	4.19	1.07	15	1
Action 27: Organize a tasting week in local schools.	3.40	1.38	14	1
Action 35: PDO partnership / Replacement service during communication actions	3.57	1.35	12	1
Action 11: Improve the physical label / graphics for the Fourme de Montbrison, PDO.	4.14	1.22	12	1
Action 02: Negotiate a minimum selling price per kilogram of Fourme de Montbrison (PDO) cheese	4.00	1.15	12	1
Action 05: Limit the PDO to exclusively raw milk	2.69	1.54	11	1
Action 12: Sell the Fourme cheese during the winter ski season.	3.71	1.47	11	1
Action 10: Promote the Fourme cheese in the traditional local restaurants.	4.23	0.84	11	1
Action 09: Promote the Fourme cheese in restaurants.	3.86	1.10	10	1
Action 23: Open a cheese bar in the PDO area.	2.79	1.42	9	1
Action 39: Improve the touristic trail, *La Route des Balcons*.	3.07	1.37	9	1
Action 49: Create a partnership grouping the PDO and local Forez products.	3.29	1.31	9	1
Action 18: Extend culinary workshops and tastings.	3.05	1.29	8	1
Action 51: Develop touristic lodging.	2.98	1.33	7	1
Action 21: New sales methods: Internet or self-service.	3.31	1.32	7	1
Action 44: Renovate the visiter halls in the dairies.	3.31	1.41	6	1
Action 16: Show product specification on the label.	2.95	1.41	4	1
Action 19: Help design installations of projects related to Fourme de Montbrison.	3.86	1.18	18	2
Action 07: Within the PDO Fourme de Montbrison, define a specific *Fourme* (blue cheese) from grass-fed herds.	3.00	1.56	15	2
Action 17: Create a cheese-maker dairy.	3.33	1.43	15	2
Action 46: Develop on-farm processing.	3.76	1.22	14	2
Action 41: Training for processing farm products.	3.71	1.15	13	2
Action 47: Hire a support technician dedicated to on-farm processing / cheese making.	3.61	1.20	10	2
Action 40: Improve the protections for the Fourme de Montbrison cheese.	3.95	1.36	9	2
Action 15: Promote the purchase of fodder outside of the PDO territory.	3.50	1.52	21	3
Action 01: Negotiate a base price for milk from dairy farmers in the PDO	4.02	0.94	19	3
Action 04: Lower and finally eliminate silage over the next 10 years	3.02	1.57	18	3
Action 20: Create a land reserve within the PDO area.	3.64	1.53	16	3
Action 14: Include the absence of pesticides and herbicides in the product specification.	3.40	1.43	16	3
Action 31: Offer financial assistance for grass producers.	3.67	1.36	16	3
Action 22: Reconsider the length of grazing time.	3.62	1.29	15	3
Action 52: Develop small windmills on the farms.	3.17	1.36	14	3
Action 36: Develop collective barn drying projects	3.00	1.53	13	3
Action 32: Organise technical information days on grazing management.	3.10	1.34	13	3
Action 37: Adapt woodlands to the PDO boundaries and the number of animals per surface area	3.61	1.32	13	3
Action 06: Prohibit the use of milking robots on farms of PDO Fourme de Montbrison	3.69	1.65	12	3
Action 50: Impose a requirement for more hay in winter feeding.	3.07	1.52	12	3
Action 43: Support agro-pastoralism in the 'Hautes-Chaumes' area (high summer pastures).	3.15	1.48	12	3
Action 38: Renovation of farm and pasture structures.	3.24	1.36	10	3

(*Continued*)

Table 2. Continued.

Proposed actions	Note/ 5[a]	Standard deviation	Participation[b]	Guidelines
Action 53: Enforce regulations for woodland zones.	3.24	1.30	10	3
Action 29: Make the dates of harvest / pasturing more flexible in the agro-environmental regulations.	3.26	1.43	9	3
Action 33: Share summer pasturing.	2.59	1.53	8	3
Action 28: Add a local breed of mountain cow to the PDO specifications.	2.37	1.40	7	3
Action 30: Assist in the regrouping and restructuring of farms.	4.07	1.16	19	4
Action 24: Establish the 'Local consultation' of the Fourme de Montbrison PDO.	3.30	1.30	13	4
Action 48: Develop jobs for livestock farmer assistance.	2.88	1.45	11	4
Action 34: Assist ownership transfer of businesses producing Fourme cheese.	3.88	1.25	10	4
Action 42: Establish a commission of farmer / cheese maker representatives at the PDO administration council.	3.12	1.25	10	4
Action 03: Increase farmer representation on the administrative board of the PDO	3.10	1.38	9	4
Action 45: Allow the consolidation of small farms to create larger farm units.	2.37	1.36	7	4
Action 25: Increase the number of PDO administrators.	2.86	1.24	6	4

[a]Each respondent was asked to rate the action from 1 to 5 (from 1 not relevant for the PDO sector to 5 extremely relevant).
[b]Number of people (farmers and dairies) wishing to get involved in the action.

refer to the expectations, needs and opinions expressed by all stakeholders in the process (Table 2). The guidelines contain three major elements.

(1) Communication focusing on the diversity of the product, reflecting the diversity of production methods and actors, and meeting the different expectations of consumers: the use of terms such as pasteurized milk, raw milk, organic milk, on-farm transformation, and artisanal production in industrial dairies.
(2) Improving product sanitary quality, in particular the safety of raw milk and the conservation of cheese for better distribution. This guideline meets the expectations of both processors and many farmers seeking empowerment and desiring to establish on-farm transformation.
(3) An orientation of dairy production towards an agro-ecological and cultural heritage approach in order to improve economic (higher milk price for farmers) and environmental development of the PDO area. This third guideline was the most discussed and led to an important reorientation of two major dimensions of current livestock systems, in the following terms:
    - The PDO's BoD proposed (i.e. not mandatory) that farmers transition to a full-hay diet for the livestock on at least 60% of PDO farms within 10 years. Full-hay diet was defined as a basic ration without fermented feed (green corn possible) for dairy cows (silage allowed for heifers). This is both an economic objective, through the valorization of milk and cheese, and an ecological objective (fewer inputs needed). The given definition is an adjustment to feed management, which was chosen to take into account the constraints of the farmers, as identified during the learning process.
    - The development of local species-rich permanent pastures in order to place an emphasis on natural heritage at the heart of dairy production. This gives

importance to the environmental value of local natural landscapes and enhances the link between grassland characteristics and the product's nutritional and sensory qualities that appeal to consumers. The resulting advantages are both economic and environmental with, for example, the development of cheese making in summer mountain pastures.

A fourth guideline concerns the internal organization of the AOP and its functioning. It questions the collective organization and the need to have the means to implement the chosen orientations.

These elements all emerged from a collective and shared ambition for regional development, that aims to perpetuate and even enhance agricultural and agri-food employment in the production area of Fourme de Montbrison, and at the same time, promote the value of the attractiveness and cultural heritage of the area. The three first guidelines are trade-offs between different sustainability dimensions, as revealed by the serious games during the third step of the support process. Such trade-offs were mainly evidenced by the La Grange® game which explicitly represents the services and impacts provided by Fourme production in this area.

Recent dynamics

During the general assembly meeting in the last step (step 6), a group of dairy farmers spontaneously came together in response to a question from the president of the PDO on the implementation of concrete actions. This group represented 30 farms, or almost half of the existing PDO farms, and all of the processors quickly joined the group. An agricultural project was then established collectively: the project"s construction occupied most of the initial process which included KS between all the stakeholders and securing funding through regional grants. With the programming of a series of nine meetings and visits, and two training sessions, the project brought the collective together around a selection of concrete actions: farm processing, barn drying, grazing, and the introduction of local breeds. Researchers helped the collective formalize operational questions and reflect on its functioning. Even if all the orientations did not lead to the implementation of actions, farm installation projects have been carried out and tests of new forage have begun. This collective group continues to meet in 2020 and has renewed its dynamics and its association with researchers through new funding obtained for three years based on the orientations defined in 2019.

Discussion

Game-based learning approach and the long-term motivation of farmers

As expected, the two serious games are the pivotal element in the support. Their use was both the culmination of the expression of the actor"s viewpoints, and the prerequisite for the formalization of a common vision. The two tools are complementary and support the passage between these two key periods of the process. They take the exchanges closer to

local expectations. Tallon, Dulcire, and Dubien (2013) have pointed out that 'troubled situations', like the initial situation of the PDO union, with the ambiguity of 'composite assemblies' can generate feelings of anxiety and unease among actors. In our study, the difficulty of negotiating and making compromises seems to have been partly overcome by the mobilization of games: The games require understanding the perspectives of the other participants, learning how to position one"s own vision, how to dialogue with others, and how to build a shared vision. The participants collectively exercised their analytical skills in this situation by proposing criteria that they consider relevant to the situation both individually and collectively. This is the case, for example, of ecosystem services provided by full-hay diet practices which valorize pastures, and which have emerged from the knowledge exchanged between farmers and partners (such as veterinarians and tourism advisors). These ecosystem services link the environmental dimensions (preservation of diversity in the Forez grasslands, input reduction, climate regulation as the result of carbon sequestration by permanent grasslands) and social dimensions (landscape aesthetic value and its links to green tourism, cheese quality) of PDO cheese production (Lamarque et al. 2014). Stakeholders thus learned that despite their different values and priorities, their knowledge is substantial and legitimate. They felt valued in the collaborative process, and they appreciated building complex knowledge together (like the cited example of ecosystem services). On this point, our study joins results previously obtained in the mobilization of games combined with others methods (Wouters et al. 2013).

In our study, despite the fact that these serious games were only played during a single day, we believe that their direct influence on the actions and actors, which continued throughout the one-year process, gives them a unique added value. Moreover, the long-term presence of the skills and knowledge acquired during the games ensure that the collective trade-offs (engineered during the games) do not disappear without consequences. The games' influence increased the motivations to collaborate and consider the others (farmers, processors) as resources. The engagement during the entire process and the construction of new projects with 30 volunteer farms (at least until 2023) makes this clear. We think that the games' long-term effect is at least in part a result of our socio-constructivist theorical basis: we felt the building of these tools should be seen as a means of helping individuals to be decentralized through socio-cognitive conflict (Johnson and Johnson 2009). Like the example of ecosystem services, this concept presumes that multiple actors faced with a problem to be solved together, will make progress even though they do not have the answer individually (Moreau, Barnaud, and Mathevet 2019). This encourages them to be more involved in the collective over the long term (Buder, Buttliere, and Cress 2017). The importance of this theoretical construction in the game seems to be an essential element that is in line with the proposals of De Smet *et al* (2014). From our perspective, this aspect is sometimes neglected in the design of games mobilized in agriculture, and the underlying agricultural concepts (agronomics, animal science, farming system, economics) are often over-emphasized to the detriment of game-based learning methods. We believe that our experience could help future game designers to overcome long-term engagement issues (Flood et al. 2018).

Game-based support to re-question and legitimise local knowledge

As pointed-out by Girard and Magda (2020), research is increasingly developing new methods to mobilize farmers' knowledge, which is seen as both expert and contextualized (Doré et al. 2011; Dogliotti et al. 2014). We think that our game-based methodology shows that it is important to recognize the value of local knowledge when developing a vision for the future. The negotiated proposal shows that local ecological knowledge can reveal new assets in transitional situations and can help resolve apparent contradictions as Klein et al. (2014) have already shown. It is also important in the subsequent definition of concrete collective actions. For example, at the end of the project, despite the clear acceptance of a position favouring the adoption of a 'full-hay diet' and the key role of this position in the 10-year strategy, farmers have very different definitions of what this actually means. The prohibition of fermented feed in the basic ration is a subject of debate that appeared from the first interviews (step 2). Some farmers prefer the simple declaration of a high percentage of hay in cattle diet without further specifications. Others would accept feed regimes that are primarily based on hay but with some added feed such as 'green' corn, and even silage for heifers. Still other farmers want a strictly 'all hay' feed regime with no other feed management options. Most farmers' arguments were purely qualitative and not based on any objective scientific or technical knowledge (Dawoe et al. 2012). The economic issue here is based on the risk of production loss in the event of drought, but there is above all an inherent fear of disruptive change associated with the confrontation between different production models: conventional, organic, sustainable (Cayre et al. 2018). In the game-based support, the perspectives and constraints of processors and craftsmen were not the only subjects of debate. The issues of farmers were also recognized and included. This was contrary to their perception of being disenfranchized prior to this participative process. The recognition of local knowledge in the proposed games seems to be one of the keys to the subsequent engagement of farmers. This is an important point in the debate over the place of local, generalized scientific, and technical knowledge in the positioning to be adopted by farm extension services (Klerkx and Leeuwis 2009; Ernst 2019; Ingram et al. 2020): games could be a way to train advisors to become facilitators in stakeholders interactions. This facilitation dimension should be fully integrated into the advisor training. Our example could be used as a case study.

Transferability to other contexts

The implementation of our game-based support in other agricultural contexts must take into account a certain number of limitations that we have identified. Firstly, as seen in the literature (Thompson et al. 2017), it is a very time-consuming process that also requires significant stakeholder (and researcher) involvement. Many meetings were necessary both in and in conjunction with the process in order to negotiate and organize. The involvement of the research team was significant (four researchers over the course of the process and up to 12 facilitators during the game day). The management of the games also requires a certain degree of training, especially on the Jeu de Territoire® which requires important supporting documents. La Grange® game is easier to conduct with a few instructions, making it easier to transfer to other contexts. The process is a

long-term investment because it should not stop with the identification of a strategy for the PDO union: At this time, we are still supporting the farmers and processors. However, our experience should not be thought of as a recipe to be applied but as the construction of a common path.

Despite the large participation in response to our support, we still faced the non-engagement of some of the farmers over the long-term. Even if this can be partially explained (age pyramid), it remains difficult to obtain the support of all farmers in the collective decision-making processes that concern them (Koutsou, Partalidou, and Ragkos 2014; Bandewar et al. 2017). In our case, this difficulty is an element that also seems related to the large size of the collective (nearly 69 farms). This suggest a possible advantage of moving towards smaller collectives by, for example, splitting up the large groups by certain themes or specific profiles, even if this reduces representativeness. Gender considerations (which we have not addressed) could also be taken into account to see if this affects farmers' engagement.

Furthermore, we noted that it was sometimes difficult to coordinate an exchange with certain actors who had very clear-cut positions on the process, particularly certain extension workers. As Landini (2020) noted, they did not have a positive perspective on the valorization of local knowledge and promoted a strictly conventional technico-economic approach. Their regular intervention on the farms thus called into question some of the benefits of the exchange and show potential risk from existing routines and relationships during the process of constructing interactions (Ingram et al. 2020).

Finally, this research reinforces the idea that taking account of local stakeholder knowledge, especially farmers, is a prerequisite to successful collaborative management in a PDO area, particularly in a contentious context. Integrating and motivating farmers to participate in decision-making is a long process that cannot be achieved in a few meetings or interviews. Our mobilization of a game-based support shows a promising and efficient methodology for improving this process. Games can re-question and legitimize local knowledge, but they require a strong theoretical anchoring to which our socio-constructivist proposal seems to address. The process is iterative and must be strongly contextualized while mobilizing different methods in coordination with the games.

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